

JIAPING XIAO

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RESEARCH INTEREST

Cyber-physical systems, machine vision, reinforcement learning, robotics and embodied intelligence.

EDUCATION

Nanyang Technological University (NTU)

Aug. 2020 - Mar. 2024

Ph.D. in Robotics and Intelligent Systems, GPA: 4.6/5.0

Asst. Prof. [Mir Feroskhan](#)

Thesis: Vision-based Learning for Drones in Collaborative and Adversarial Environments

Beihang University (BUAA)

Sep. 2014 - Mar. 2017

M.S. in Flight Dynamics and Control, GPA: 90.3/100 (1/31)

Prof. Zuojun Shen

Thesis: Entry Guidance Method under Dynamic Atmosphere and Uncertainty Evaluation

Beihang University (BUAA)

Sep. 2010 - Jul. 2014

B.E. in Aircraft Design and Engineering, GPA: 86.2/100 (73/266)

Prof. Zuojun Shen, Prof. Hu Liu

Project: Uncertainties Modeling and Simulation for Flight Vehicles

HONORS AND AWARDS

Technological Innovation: **First-class prize** in 21st “Feng Ru Cup” (2011), **Third-class prize** in 23rd “Feng Ru Cup” (2013), **Excellent project** in the 6th Student Research Training Program(SRTP)(2013), **Excellent project** in the 8th Student Research Training Program (SRTP) (2014), **Gold medal** in [7th China International College Students’ “Internet+” Innovation & Entrepreneurship Competition](#) (2021), **Excellent prize** in [The “Chunhui Cup” Oversea Students Innovation and Entrepreneurship Competition](#) (2021), **4th-place** in [Singapore Amazing Flying Machine Competition \(SAFMC\) Category E](#) (2024), **5th-place** in [CVPR UG2+ Challenge 2024 Track 3](#) (2024), Certificate of Merit in [IEEE SMC Society Open-Source Competition 2025](#) (2025).

Social Work: **Outstanding student leader** award (2011), **Advanced individual** in social practice (2013), **Excellent student award** in international graduate summer school in aeronautics and astronautics (2015), **Outstanding contribution** in NTU GSA (2022-2023).

Academic Award: Third Prize Academic Scholarship (B.E.), **First Prize** Academic Scholarship (M.S.), Excellent Graduate Student Award (2017), [Ross Fellowship](#) (Purdue University, 2018), Nanyang Research Scholarship (NTU, 2020), [Schmidt Science Fellows](#) Nomination (1/400 global, Cohort 2025), 2023 Chinese Government Award for Outstanding Self-financed Students Abroad (1/650 global, 2024).

PUBLICATIONS

JOURNAL

38. [IOTJ] **J. Xiao**, X. Fang, Q. Jia, M. Feroskhan. “Learning Resilient Formation Control of Drones with Graph Attention Network.” *IEEE Internet of Things Journal*, 2025. [\[pdf\]](#) (IF: 8.2, JCR-Q1)
37. [TNNLS] **J. Xiao**, R. Zhang, Y. Zhang, M. Feroskhan. “Vision-based Learning for Drones: A Survey.” *IEEE Transactions on Neural Networks and Learning Systems*, 2025. [\[pdf\]](#) (IF: 10.2, JCR-Q1)

36. [TRO] L. Chen[†], **J. Xiao[†]**, Y. Zheng, N. A. Alagappan, M. Feroskhan. “Design, Modeling, and Control of a Coaxial Drone.” *IEEE Transactions on Robotics*, 2024.[pdf] (IF: 9.4, JCR-Q1, Co-first author)
35. [AIS] **J. Xiao**, P. Pisutsin, M. Feroskhan. “Toward Collaborative Multitarget Search and Navigation with Attention-Enhanced Local Observation.” *Advanced Intelligent Systems*, 2024.[pdf] (IF: 6.8, JCR-Q1)
34. [TAI] **J. Xiao**, M. Feroskhan. “Learning Multi-Pursuit Evasion for Safe Targeted Navigation of Drones.” *IEEE Transactions on Artificial Intelligence*, 2024.[pdf] (IF: 7.7)
33. [TIV] **J. Xiao**, J. H. Chee, M. Feroskhan. “Real-Time Multi-Drone Detection and Tracking for Pursuit-Evasion With Parameter Search.” *IEEE Transactions on Intelligent Vehicles*, 2024.[pdf] (IF: 14.0, JCR-Q1)
32. [TNNLS] **J. Xiao**, P. Pisutsin, M. Feroskhan. “Collaborative Target Search With a Visual Drone Swarm: An Adaptive Curriculum Embedded Multistage Reinforcement Learning Approach.” *IEEE Transactions on Neural Networks and Learning Systems*, 2023.[pdf] (IF: 10.2, JCR-Q1)
31. [TVT] **J. Xiao**, M. Feroskhan. “Cyber attack detection and isolation for a quadrotor uav with modified sliding innovation sequences.” *IEEE Transactions on Vehicular Technology*, 2022.[pdf] (IF: 6.1, JCR-Q1)
30. [TIV] L. Chen[†], **J. Xiao[†]**, C. Teo, M. Feroskhan. “Air-Ground Collaborative Control for Angle-Specified Heterogeneous Formations.” *IEEE Transactions on Intelligent Vehicles*, 2024.[pdf] (IF: 14.0, JCR-Q1, Co-first author)
29. [AA] Q. Jia, **J. Xiao***, L. Bai, Y. Zhang, R. Zhang, M. Feroskhan. “Space situational awareness systems: Bridging traditional methods and artificial intelligence.” *Acta Astronautica*, 2024.[pdf] (IF: 3.1, JCR-Q1, Corresponding author)
28. [RAL] Y. Zhang, **J. Xiao***, M. Feroskhan. “Learning Cross-Modal Visuomotor Policies for Autonomous Drone Navigation.” *IEEE Robotics and Automation Letters*, 2024.[pdf] (IF: 4.6, JCR-Q1, Corresponding author)
27. [TII] Q. Jia, **J. Xiao**, M. Feroskhan. “Multi-Target Assignment under Uncertain Information Through Decision Support Systems.” *IEEE Transactions on Industrial Informatics*, 2024.[pdf] (IF: 11.7, JCR-Q1)
26. [TAI] Y. Zhang, C. Yan, **J. Xiao**, M. Feroskhan. “NPE-DRL: Enhancing Perception Constrained Obstacle Avoidance with Non-Expert Policy Guided Reinforcement Learning.” *IEEE Transactions on Artificial Intelligence*, 2024.[pdf] (IF: 7.7)
25. [MIR] Y. Chen, **J. Xiao***. “Target Search and Navigation in Heterogeneous Robot Systems with Deep Reinforcement Learning.” *Machine Intelligence Research*, 2024.[pdf] (IF: 6.4, JCR-Q1, Corresponding author)
24. [TCST] L. Chen, **J. Xiao**, R.C.H. Lin, M. Feroskhan. “Angle-Constrained Formation Maneuvering of Unmanned Aerial Vehicles.” *IEEE Transactions on Control Systems Technology*, 2023.[pdf] (IF: 4.9, JCR-Q1)
23. [CSA] J. Jiang, **J. Xiao***, K. Tang. “Data-Driven Cyber-Physical System Security Threat Analysis Model and Method for Smart Cities.” *Computer Systems and Applications*, 2018.[pdf] (IF: 2.08, Corresponding author)
22. [CTA] **J. Xiao**, J. Jiang, C. She. “Data attack detection for an unmanned aerial vehicle control system using innovation sequences.” *Control Theory and Applications*, 2017.[pdf] (IF: 0.80)

CONFERENCE

21. [CVPR] Y. Chen, J. Gao, Y. Pan, S. Song, Z. Zhang, J. Xiao*, M. Feroskhan. “GeniNav: Generative Model Driven Image-Goal Navigation via Imagination-Guided Consistency Flow Matching.” *Conference on Computer Vision and Pattern Recognition 2026 (CVPR-26)*, Denver, U.S., 2026. (**25% acceptance rate**, CCF-A, Corresponding author)
20. [IJCAI] **J. Xiao[†]**, R. Zhang[†], Y. Zhang, Q. Jia, L. Bai, M. Feroskhan. “SpaceDet: a large-scale space-based image dataset and RSO detection for space situational awareness.” *The 34th International Joint Conference on Artificial Intelligence (IJCAI-25)*, Montreal, Canada, 2025. (**17% acceptance rate**, CCF-A, Co-first author)

19. [ADMA] X. Zhou, J. Zhou, **J. Xiao**, Y. Zhang, X. Hao, J. Wang, B. Chen, Q. Wen. “RPGCN: Relational Probabilistic Graphs for EEG-based Emotion Mining.” *21st International Conference on Advanced Data Mining and Applications, Japan*, 2025. (CCF-C, **Best paper award**)
18. [COLING] Z. Li, S. Zheng, **J. Xiao**, X. Li, L. Wang. “UCTG: A Unified Controllable Text Generation Framework for Query Auto-Completion.” *The 31st International Conference on Computational Linguistics*, 2025.[\[pdf\]](#) (CCF-B)
17. [CASE] P. Pisutsin[†], **J. Xiao**[†], M. Feroskhan. “Omnidrone-Det: Omnidirectional 3D Drone Detection in Flight.” *2024 IEEE 20th International Conference on Automation Science and Engineering*, 2024.[\[pdf\]](#) (Co-first author)
16. [IJCAI] C. Liu, X. Zhou, **J. Xiao**, Z. Zhu, L. Zhai, Z. Jia, Y. Liu. “VSGT: Variational Spatial and Gaussian Temporal Graph Models for EEG-based Emotion Recognition.” *The 33rd International Joint Conference on Artificial Intelligence (IJCAI-24)*, Jeju, Korea, 2024.[\[pdf\]](#) (CCF-A)
15. [CAI] Y. Zhang, R. Zhang, Q. Jia, **J. Xiao**, L. Bai, M. Feroskhan. “Astro-Det: Resident Space Object Detection for Space Situational Awareness.” *2024 IEEE Conference on Artificial Intelligence*, 2024.[\[pdf\]](#)
14. [EMBC] X. Zhou, D. Lin, Z. Jia, **J. Xiao**, C. Liu, L. Zhai, Y. Liu. “An EEG Channel Selection Framework for Driver Drowsiness Detection via Interpretability Guidance.” *2023 45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Sydney, Australia, 2023.[\[pdf\]](#)
13. [CAI] **J. Xiao**, YXM. Tan, X. Zhou, M. Feroskhan. “Learning Collaborative Multi-Target Search for A Visual Drone Swarm.” *2023 IEEE Conference on Artificial Intelligence*, 2023.[\[pdf\]](#)
12. [CAI] X. Zhou, C. Liu, **J. Xiao**^{*} and Y. Liu^{*}. “EEG-based Sleep Staging with Hybrid Attention.” *2023 IEEE Conference on Artificial Intelligence*, 2023.[\[pdf\]](#) (Corresponding author)
11. [WCICA] **J. Xiao**, J. Jiang. “Real-time Security Evaluation for Unmanned Aircraft Systems under Data-driven Attacks.” *2018 13th World Congress on Intelligent Control and Automation*, 2018.[\[pdf\]](#)
10. [CGNCC] **J. Xiao**, Z. Shen. “Uncertainty analysis of entry guidance based on confidence interval estimation.” *2016 IEEE Chinese Guidance, Navigation and Control Conference*, 2016.[\[pdf\]](#)
9. [CCC] **J. Xiao**, Z. Shen. “Stability analysis of Trajectory Tracking Entry Guidance based on singular perturbation theory.” *2016 35th Chinese Control Conference*, 2016.[\[pdf\]](#)

UNDER REVIEW

8. [TITS] **J. Xiao**, C. W. Tsao, Y. Zhang, M. Feroskhan. “Foundation Model Guided Path Planning for Intent-Aware Autonomous UAV Navigation.” *IEEE Transactions on Intelligent Transportation Systems*, 2025. (Under Review)
7. [TRO] **J. Xiao**, X. Cao, M. Feroskhan. “Collaborative Visual Multitarget Search with Heterogeneous Graph Attention Embedded Multistage Learning.” *IEEE Transactions on Robotics*, 2025. (Under Review)
6. [TAES] **J. Xiao**, L. Bai, M. Feroskhan. “Hybrid Physics-Informed Neural Orbit Propagation Under Sparse Space-Based Observations.” *IEEE Transactions on Aerospace and Electronic Systems*, 2025. (Major revision)
5. [TPAMI] Y. Zhang, **J. Xiao**^{*}, C. Yan, M. Feroskhan. “Oracle-Guided Masked Contrastive Reinforcement Learning for Visuomotor Policies.” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.(IF: 20.8, Under review, Corresponding author)
4. [TPAMI] Y. Zhang, C. Yan, J. Yu, **J. Xiao**^{*}, M. Feroskhan. “Learning Adaptive Cross-Embodiment Visuomotor Policy with Contrastive Prompt Orchestration.” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.(IF: 20.8, Under review, Corresponding author)
3. [TII] M. Li, **J. Xiao**^{*}, M. Feroskhan. “Learning Multi-Drone Coverage Path Planning with Hierarchical Observation Enhanced Graph Attention Network.” *IEEE Transactions on Industrial Informatics*, 2025. (IF: 11.7,

Major revision, Corresponding author)

2. [IOTJ] Y. Zhang, H. Yu, **J. Xiao***, M. Feroskhan. “VLFly: Grounded Vision-Language Navigation for AAVs with Open-Vocabulary Goal Understanding.” *IEEE Internet Of Things Journal*, 2025.(IF: 8.2, Minor revision, Corresponding author)
1. [TSMC] L. Bai, **J. Xiao***, R. Zhang, M. Feroskhan. “Angles-Only Initial Orbit Determination with Sinusoidal-Aware Neural Network for Enhanced Space Situational Awareness.” *IEEE Transactions on Systems Man and Cybernetics-Systems*, 2025. (IF: 8.7, Major revision, Corresponding author)

PATENTS

- A Novel Design And Controller Of A Coaxial Drone, Inventors: FEROSKHAN Mir Alikhan Bin Mohammad, CHEN Liangming, **XIAO Jiaping**, ZHENG Yumin, ALAGAPPAN N Arun, *US 19/048,519*
- A Novel Perception System For Space-Based SSA, Inventors: **XIAO Jiaping**, Mir Alikhan Bin Mohammad FEROSKHAN, ZHANG Rangya, BAI Lu, *2025-003-01-SG PRV*
- A Novel Physical Inspired Learning Approach To Angles-Only Orbit Determination For Space-Based SSA, BAI Lu, Mir Alikhan Bin Mohammad FEROSKHAN, **XIAO Jiaping**, ZHANG Rangya, *2025-302-01-SG PRV*

PROJECTS

- “**Stalker: Sensor-based Following of Multiple Targets**”, (*Project Leader*), Jul. 2025-now, Supported by Temasek Laboratories, Singapore.
- “**Aeromaze: Aerial Swarm Autonomous Navigation Challenge in GPS-denied Environment**”, (*Core member*), Jul. 2025-Nov. 2025, Supported by Temasek Laboratories, Singapore.
- “**Development of an onboard, mobile vision-based SASS**”, (*Acting Project Leader*), Jun. 2023-Aug. 2025, Supported by Economic Development Board (EDB), Singapore.
- “**ParcelDrone-A Modular, Self-powered Multirotor Approach to Aerial Parcel Delivery**”, (*Core Member*), Apr. 2022-Mar. 2025, Supported by A*STAR, Singapore.
- “**Research on Parallel Test Theory and Methods**”, (*Principal Investigator*), 2019-2020, Supported by Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences.
- “**Three-dimensional Entry Guidance and Simulation under Complex Atmosphere**”, (*Project Leader*), Oct. 2014-May. 2015, Supported by School of Aeronautical Science and Engineering, BUAA.
- “**Design of Fly-Fish-like Hybrid Power Duct UAV**”, Nov. 2012-May. 2013, (*First Author*), Supported by The 8th Student Research Training Program (SRTP), BUAA.

EXPERIENCES

Nanyang Technological University (NTU) Mar 2024-Now
Research Fellow

- Conduct high impact innovative research and research collaboration.

NTU Graduate Student Associate (GSA) Apr 2022-Sep 2023
Academic Programs Director

- Lead the academic committee of NTU GSA.

Nanyang Technological University (NTU) Jan 2022-Now
Teaching Assistant

- Undergrad Lab sessions “**Introduction to Aerodynamics**” (2023 Spring).
- Undergrad course “**Air Traffic Management and Operations**” (2022 Fall / 2023 Fall).
- Undergrad course “**Autonomous Flight**” (2025 Spring / 2026 Spring).
- Ph.D. course “**GP8000 PGR ICC Artificial Intelligence Literacy**” (2025 Spring / 2025 Fall).

National University of Singapore (NUS)

Jan 2021-Jun 2021

Ph.D. Exchange Student

- NTU-NUS Ph.D. Exchange Program, ECE.

Institute of Software, Chinese Academy of Sciences (ISCAS)

Mar 2017-Jul 2020

Engineer

- Responsible for the modeling and algorithm in the game simulation system;
- Research about the SPO-GA/DRL algorithms applied in the tasks schedule and gaming system.

Purdue University in West Lafayette

Aug 2018-Dec 2018

Ph.D. Student

- College of Engineering with fully funded Ross Fellowship, focusing on convex optimization and secure cyber-physical systems (Supervisor: Inseok Hwang).

Fujitsu

Jun 2015-Sep 2015

Algorithm Engineer Intern

- Responsible for the research and development of indoor positioning and navigation algorithms using step detection and WIFI signal computation.

Microsoft

Mar 2014-Jul 2014

Technical Support Intern

- Responsible for the company’s network and database maintenance, troubleshooting and hardware maintenance.

ACADEMIC SERVICES

- Guest Editor of [Electronics](#) (IF: 2.6).
- Editorial Board Member of [Exploration](#) (IF: 22.5).
- Editorial Board Member of [AI and Autonomous Systems](#).
- Editorial Board Member of [Robot Learning](#).
- Editorial Board Member of [Journal of Artificial Intelligence & Control Systems](#).
- The Intelligent Systems Applications Technical Committee ([ISATC](#)) Member of IEEE Computational Intelligence Society
- Session Chair of the International Joint Conference on Artificial Intelligence ([IJCAI](#)) 2025
- Teaching Assistant at NTU.
- Tutor for GP8000 PGR ICC Artificial Intelligence Literacy course.
- **Workshop Chair** in the 2nd/3rd International conference on the Frontiers of Intelligent Control, Measurement and Signal Systems ([FICMSS 2025/2026](#)).
- **Registration Chair** in the 2026 IEEE International Conference on Human-Machine Systems ([ICHMS 2025](#)).
- **International Programme Committee Member** in the International Conference on Drones and Unmanned Systems 2025 ([DAUS’ 2025](#)).

- **Technical Program Committee** in the 9th International Conference on Computer Science and Application Engineering ([CSAE 2025](#)).
- Journal Reviewer: *IEEE Transactions on Robotics (TRO)*, *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, *IEEE Transactions on Cybernetics (TCYB)*, *IEEE Transactions on Intelligent Vehicles (TIV)*, *IEEE Transactions on Vehicular Technology (TVT)*, *IEEE Transactions on Control Systems Technology (TCST)*, *IEEE Transactions on Aerospace and Electronic Systems (TAES)*, *IEEE Transactions on Artificial Intelligence (TAI)*, *IEEE Transactions on Automation Science and Engineering (TASE)*, *IEEE Robotics and Automation Letters (RAL)*, *IEEE Sensors Journal*, *Neurocomputing*, *Advanced Intelligent Systems (AIS)*, *Engineering Applications of Artificial Intelligence (EAAI)*, *Scientific Report*, *The Journal of Engineering*.
- Conference Reviewer: The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025), 2024 IEEE 20th International Conference on Automation Science and Engineering (CASE 2024), ICLR 2025, CVPR 2026, ECCV 2026.
- IEEE Member.

INVITED TALKS

4. “Vision-based Learning for Multiagent Systems”, *NTU AI Meetup 2026*, Singapore, 2025.
3. “Vision-based Learning for Intelligent Systems: Methods and Application”, *SUSTech*, Shenzhen, China, 2025.
2. “LetsTalkResearch Workshop 2024”, *NTU*, Singapore, 2024.
1. “Leadership in NTU”, *NTU*, Singapore, 2023.

MENTORING, LEADERSHIP & ACTIVITIES

- Mentor for Ph.D. Student NTU (4 students), Master Student NTU (4 students), FYP NTU (15 students), URECA NTU (2 students), CN Young NTU (1 student)
- Academic Program Director of NTU GSA (2022-2023)
- Academic Member of NTU GSA (2021-2022)

SKILLS

- **Programming Languages:** NCRE-2(C++), Linux C/C++, C#, Python, MATLAB
- **Software:** OpenCV, Pytorch, ROS, Pixhawk 4, Gazebo, Unity
- **Hardware:** RaspberryPi, NVIDIA Jetson
- **Languages:** English, Chinese

REFEREES

- Asst. Prof. [Mir Feroskhan](#) (Supervisor), Email: mir.feroskhan@ntu.edu.sg
- Assoc. Prof. [Chen Lyu](#) (Thesis Advisory Committee Member), Email: lyuchen@ntu.edu.sg
- Assoc. Prof. [Sameer Alam](#) (Teaching Assistant Mentor), Email: sameeralam@ntu.edu.sg
- Assoc. Prof. [Bihan Wen](#) (Thesis Advisory Committee Member), Email: bihan.wen@ntu.edu.sg